ITMO UNIVERSITY

COURSE PREVIEW · ITMO UNIVERSITY · 16 SEPTEMBER 2026

Social Media Analytics, Agents, and LLMs in Marketing and Business

A teaser walkthrough of the course — one topic per lecture.

Prof. Aleks Farseev · Daniel Onishchenko · Jack Zhao

1. Intro
2. Profiling
3. Networks
4. Retrieval
5. RecSys
6. Metrics
7. LLMs
8. Agents
9. Marketing
10. Business

★ **Project defence**

YOUR FACILITATORS

Research, industry and teaching — in one room.



Prof. Aleks Farseev

COURSE LEAD

Founder & CEO, SOMIN.AI. PhD, National University of Singapore. 30+ papers at AAAI, ACM Multimedia, SIGIR, TOIS.



Daniel Onishchenko

FACILITATOR

DEFY Research. Lead author of EDRA, self-improving LLM outreach agents (ACM Multimedia 2026 demo).



Jack Zhao

FACILITATOR

ITMO University. Co-author of EDRA (ACM Multimedia 2026 demo).

WHERE THIS COURSE LEADS · ALUMNI

Two graduates of this course. Two of the most competitive AI teams in the world.



Denis Mekhanikov

Senior Software Engineer

Google DeepMind · London

Google DeepMind

- ✓ **From ITMO to DeepMind.** Graduate of this course and PhD student of Prof. Farseev.
- ✓ **DeepMind** is Google's AI lab — the team behind Gemini and AlphaFold.



Qi Yang

Lead ML Engineer, Marketing & Advertising

Keeta · Meituan

Keeta

- ✓ **From robotics to ad AI.** PhD student of Prof. Farseev on social media user profiling; former tutor of this course.
- ✓ **Keeta** is Meituan's global food-delivery app: Hong Kong, Saudi Arabia, Qatar, Kuwait, UAE — and Brazil next.

Their foundations — user profiling, recommendation, machine learning in production — are the lectures on your list.

SOMIN · GOOGLE CLOUD · GOOGLE DEEPMIND

The research behind this course is on Google DeepMind's radar.

Google DeepMind

> **SOMIN's Research Department has consistently aligned with Google's AI advancements, leading to cross-references by Google DeepMind in next-generation advertising systems such as ACAI.**

Google Cloud customer story — cloud.google.com/customers/somin-ai

ACAI is Google's AI co-creation system for small-business advertising — you'll take it apart in Lecture 9.

BUILT ON GEMINI & GOOGLE CLOUD

12×

FASTER

campaign planning — weeks down to 13 minutes per brief

78%

ACCURACY

predicting click-through rates — 3× more accurate than average human estimates

3 hrs

TO INSIGHT

cost-per-click insights, against a 24-hour SLA

SOURCE Google Cloud, “SOMIN” customer story — cloud.google.com/customers/somin-ai (figures and quote as published by Google Cloud).

THE COURSE IN ONE PICTURE

Data about people → systems that decide what to show → AI that acts → business value.

01

UNDERSTAND

Who is the user?

- 1 Intro to social media analytics
- 2 User profiling
- 3 Networks, influence & clicks

02

MATCH

What should they see?

- 4 Information retrieval
- 5 Recommendation systems
- 6 Evaluation metrics

03

GENERATE & ACT

Who does the work?

- 7 Large language models
- 8 Agentic AI

04

APPLY

What is it worth?

- 9 Marketing applications
- 10 Business applications

★ **FINAL PROJECT DEFENCE** Everything above, built into one tech-enabled product by your team.

COURSEWORK · WHAT YOU'LL BUILD

Two labs to learn the tools. One mini start-up to prove it.

01

LAB 1

Practical machine learning

Individual. A hands-on ML task on real social media data — for example, classifying microblog posts by topic or sentiment.

02

LAB 2

Practical machine learning

Individual. A harder, multi-source task — for example, predicting user personality from text, images and behaviour.

03

FINAL PROJECT

A mini start-up

Teams of 2–3. Build a working, tech-enabled application with a real use case, using methods from the course. Checkpoints with the facilitators along the way.

04

EXAM

The project defence

Present your product and results, run a live demo, then defend your technical choices in a Q&A with the facilitators.

WHAT THE PROJECT IS JUDGED ON

Idea & design · Technology · Implementation, demo & report · Going the extra mile

HOW YOU'RE GRADED · NUS GRADING SYSTEM

Grades are earned by doing the work — on time.

20%

ATTENDANCE

Points for showing up and taking part in lectures and discussions.

30%

TWO LABS

Points for each lab — and for handing it in on time.

50%

PROJECT + DEFENCE

Judged on the product you build and how you defend it.

THE RULES

- ✓ **Late = lost points.** Every late submission costs marks.
- ✓ **Knowing isn't enough.** Do nothing all semester and even knowing it all barely earns a C.
- ✓ **Relative grading.** Final letter grades follow the NUS system.

NUS LETTER GRADES

A+	← steady work, on time
A	
A−	
B+	
B	
B−	
C+	
C	**← no coursework, knows it all**
D+	
D	
F	

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01

LECTURE 1

Introduction to social media analytics

THE PLATFORM DILEMMA

> **Twitter's users are its content. Show them too many ads, and they post less — and the platform loses its value.**

From the ITMO course. Every social platform balances attention, content and revenue. Analytics is how they keep that balance — and how everyone else learns from their data.

LECTURE 1 · WHAT YOU'LL LEARN

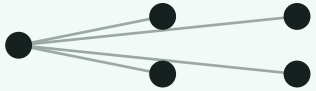
Everyone is a publisher now — and every post is data.



ONE → ONE

Communication media

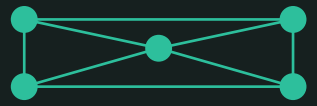
Phone, email



ONE → MANY

Broadcast media

Filter, then publish



MANY → MANY

Social media

Publish, then filter

TODAY'S SOCIAL DATA IS LIVE, BIG AND 3M

MULTI-SOURCE

Posts, check-ins, search, e-commerce, IoT sensors

MULTIMEDIA

The newest platforms are image- and video-first

MULTILINGUAL

Informal, abbreviated, changing vocabulary

THE CHALLENGES WE'LL TACKLE

>30%

of user-generated content is noise, spam or rumours.

- ✓ **Users:** influence, communities, profiles
- ✓ **Content:** unstructured, noisy, huge
- ✓ **Models:** privacy, online + offline data

SOURCE Farseev, ITMO Social Media Analytics — Lecture 1: Introduction (social media platforms, observatories and challenges).

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02

LECTURE 2

Social media user profiling

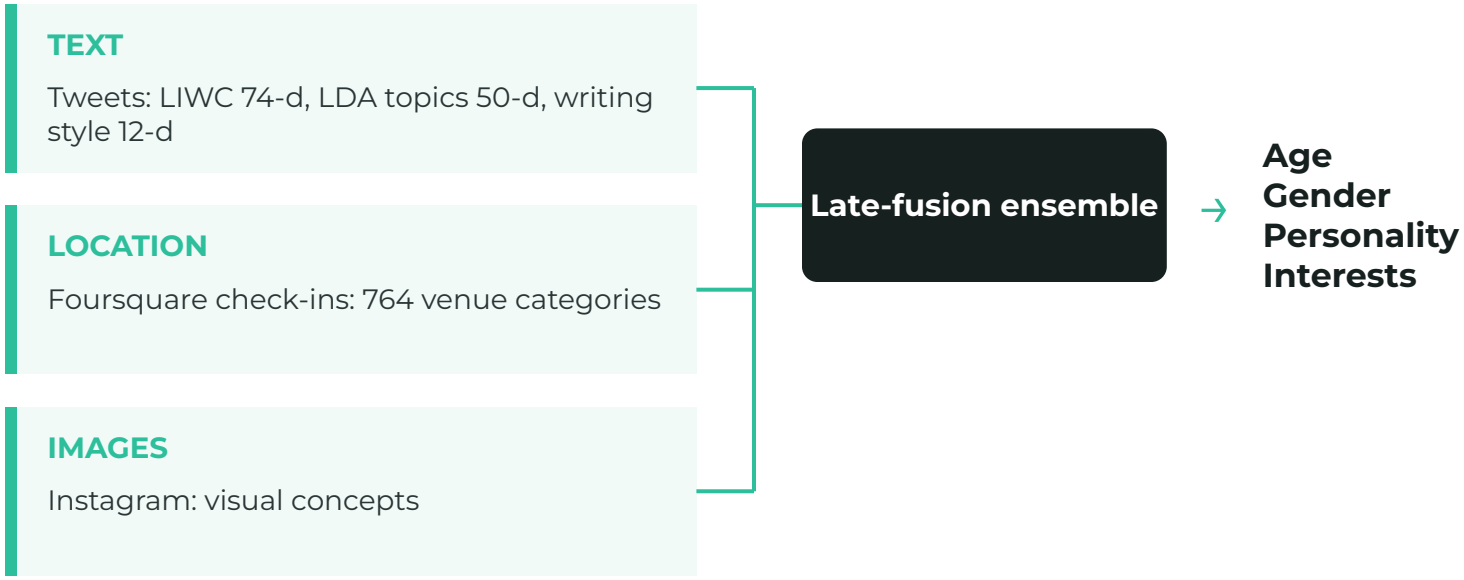
A PROFILE BUILT FROM PUBLIC POSTS

> **Tends to stay at home, visits local pubs and a shopping mall daily… Advertise a new beer brand and new car models.**

An inferred profile from the ITMO course slides. How much do your tweets, check-ins and photos give away — and should anyone be allowed to act on it?

LECTURE 2 · WHAT YOU'LL LEARN

Each platform sees one side of you. Fusion sees the whole.



165k tweets · 2k check-ins · 4k Instagram images · 305 users

WHAT THE DATA SHOWED

- ✓ **Images** are useful for gender prediction
- ✓ **Locations** are useful for age prediction
- ✓ **Text** helps with both
- ✓ **Ethics:** consent, privacy and profiling at scale

SOURCE Farseev, ITMO Social Media Analytics — Multi-Source Data Fusion; LBSN, Mobility & Demography; Farseev et al., ICMR 2015.

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03

LECTURE 3

Social networks, influence & clicks

THE STRENGTH OF WEAK TIES

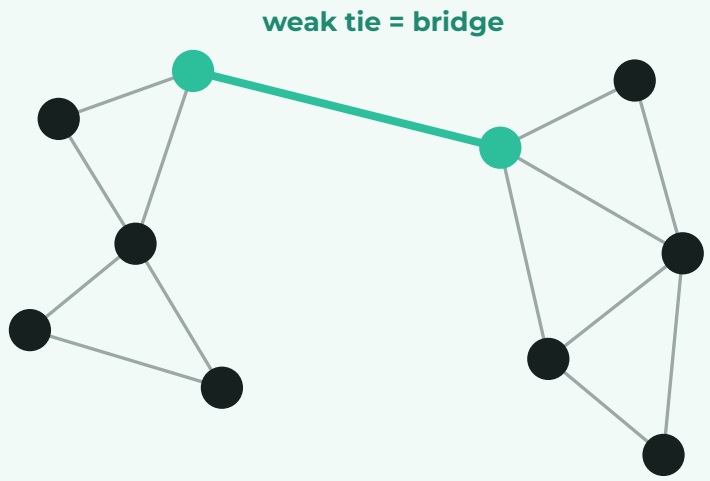
> **You're more likely to hear about your next job from an acquaintance than from a close friend.**

Granovetter's classic finding: weak ties bridge to new information. Who you follow, who follows you, and how you click shape what spreads — and expose who is fake.

LECTURE 3 · WHAT YOU'LL LEARN

How you connect and click says as much as what you post.

TWO COMMUNITIES, ONE BRIDGE



Small worlds · clustering · triadic closure · homophily

INFLUENCE

Followers ≠ influence

Influence varies by topic. TwitterRank finds who matters for which topic. In a 6,748-user Singapore study, 72.4% of users follow back over 80% of their followers.

VIRALITY

What makes content spread

Popularity depends on content, user, network and timing — and timing features predict best. Breadth of sharing matters more than depth.

CLICKS

Catching fake accounts

Clickstream patterns expose Sybil accounts: 16K users, 6.8M clicks, under 0.7% false positives. A scan of 1M users flagged 22K suspicious accounts.

SOURCE ITMO SMA — Graph Theory & Social Networks (after Easley & Kleinberg); Event Detection & Predictive Analytics (TwitterRank; Cheng et al., WWW 2014); Spam & Sybil Detection.

04

LECTURE 4

Information retrieval systems

THE SCALE OF THE PROBLEM

> **Google indexes an estimated 35 trillion pages — perhaps 4% of the information on the Internet.**

Figure as cited in the ITMO course. So how does any system find the ten results you actually want, in milliseconds? And why does the same machinery now sit underneath every LLM chatbot?

LECTURE 4 · WHAT YOU'LL LEARN

From a pile of documents to a ranked list in five steps.

01

PROCESS

Tokenise, remove stopwords, stem

02

INDEX

Inverted index: term → documents

03

WEIGHT

tf · idf — rare terms matter more

04

RANK

Cosine similarity, BM25

05

EVALUATE

Precision, recall, MAP (Lecture 6)

WORKED EXAMPLE · QUERY: “information retrieval”

D1	information retrieved by VS retrieval methods	**sim = 3**
D2	information theory forms the basis for probabilistic methods	**sim = 1**
D3	He retrieved his money from the safe	**sim = 1**

THE KEY FORMULAS

$$w = (1 + \log tf) \times \log(N / df)$$

$$\cos(q, d) = q \cdot d / (|q| \, |d|)$$

“idf helps precision, tf helps recall.”

SOURCE ITMO SMA — Text Analysis & Classification; Nayak & Raghavan, Stanford CS276 Lecture 6 — web.stanford.edu/class/cs276/19handouts/lecture6-tfidf-1per.pdf

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05

LECTURE 5

Recommendation systems

THE PREDICTION PROBLEM

She checks in at a flower shop on a Saturday morning. Where will she go next?

One right answer among thousands of places, and almost no data per user. That is recommendation: extreme sparsity, heavy class imbalance — and a business that depends on getting it right.

LECTURE 5 · WHAT YOU'LL LEARN

Search answers a query. Recommenders guess what you'd ask for.

THE UTILITY MATRIX

	Avatar	LOTR	Matrix	Pirates
Alice	4		5	1
Bob	5	5	4	
Carol		?		4
David			3	?

Blank = unknown. The job is to fill the ? cells — including for brand-new users (cold start).

HOW TO FILL THEM

- ✓ **Popularity:** what most people like
- ✓ **Content-based:** items like the ones you liked
- ✓ **Collaborative filtering:** people like you liked it
- ✓ **Matrix factorisation:** M ≈ P·Q, hidden tastes

IN THE WILD

35M

Foursquare check-ins used to predict users' next place.

Combining signals — popularity, friends, venue type, distance — beats any single one.

Item-item CF: $\hat{r} = \Sigma s_i \cdot r \square / \Sigma s_i \square$

SOURCE ITMO SMA — LBSN, Mobility & Demography (Noulas et al., ICDM 2012); Multi-Source Data Fusion; Leskovec, Stanford CS246 “Recommender Systems I”.

06

LECTURE 6

Evaluation metrics

A MILLION-DOLLAR DECIMAL

> **Netflix offered $1,000,000 to anyone who could cut its recommendation error by 10%.**

Its own system scored an RMSE of 0.9514. But is a lower error what viewers actually want? Choosing the metric is choosing what “good” means.

LECTURE 6 · WHAT YOU'LL LEARN

Choose the metric, and you've chosen what “good” means.

PRECISION VS RECALL

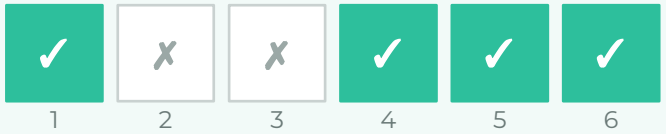
	Relevant	Not
Retrieved	a	b
Missed	c	

P = a / (a + b)

R = a / (a + c)

F1 = 2PR / (P + R)

AVERAGE PRECISION



AveP = (1 + 2/4 + 3/5 + 4/6) / 4 = 0.69

Precision at each relevant result, averaged. MAP = the mean over many queries. Order matters: the same four hits ranked first would score 1.0.

WHICH METRIC, WHEN?

- **RMSE:** predicted ratings
- **Precision@k:** top-k lists
- **MAP, NDCG:** full rankings
- **Micro vs macro F1:** common vs rare classes
- **Beyond accuracy:** diversity, novelty, coverage

SOURCE ITMO SMA — Text Analysis & Classification (evaluation measures); Leskovec, Stanford CS246 “Recommender Systems I” (Netflix Prize, RMSE, precision at top 10).

07

LECTURE 7

Introduction to large language models

FROM THE FUTURE-PROOF FORUM '26 STAGE

> **AI is still a statistical model. Given a prompt, it returns the most likely response — not the most original one.**

Feed the same prompt to the same model and most brands in a category land on the same answer: the “sea of sameness”. To use LLMs well, you need to know what they actually are.

LECTURE 7 · WHAT YOU'LL LEARN

An LLM predicts the next token. Everything else is how it's trained.

A LANGUAGE MODEL SCORES SEQUENCES

P(the mouse ate the cheese) = 0.02

P(the cheese ate the mouse) = 0.001

IN PRACTICE · GOOGLE CLOUD CASE STUDY

12× faster campaign planning when an LLM (Gemini) is grounded in real market data — weeks of work down to ~13 minutes per brief.

TOKENISE
Byte-pair encoding

→

PRE-TRAIN
Predict next token on web-scale text

→

SFT
Few thousand instruction examples

→

RLHF / DPO
Align with human preferences

Pre-training gives you GPT-3. Post-training (SFT → RLHF) gives you ChatGPT.

SOURCE Y. Dubois, Stanford CS229 guest lecture “Introduction to Building LLMs”, 2024; Farseev, Future-Proof Forum '26, Google Manila; cloud.google.com/customers/somin-ai

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08

LECTURE 8

Agentic AI systems

“CAN I GET A REFUND FOR MY ORDER?”

“

A chatbot quotes the refund policy. An agent looks it up, asks for your order number, calls the order system and confirms the refund.

Same LLM underneath. The difference is planning, tools and memory — and the risk that comes with letting software take actions on a business's behalf.

LECTURE 8 · WHAT YOU'LL LEARN

Agents plan, use tools — and can rewrite their own rules.

THE AUTONOMY LADDER

Agent = LLM + tools + memory + planning

- **Multi-agent**
- **Agentic workflow**
- **RAG**
- **Chain**
- **Prompt**

CASE STUDY · EDRA, ACM MULTIMEDIA 2026 ↺ loop

1 → **Cluster** visitor profiles

2 → **Induce** a rule per segment

3 → **Apply** nearest-neighbour rule

4 **Revise** LLM rewrites weak rules

ESTIMATED REWARD · 744 REAL OUTREACH INTERACTIONS



SOURCE K. Katanforoosh, Stanford CS230 Lecture 8 “Agents, Prompts, and RAG”, Autumn 2025; Onishchenko, Haberzettl, Makarov, Zhao, Cassidy, Farseev, “EDRA”, ACM MM '26 demo.

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09

LECTURE 9

Marketing applications

BURGER KING HAD A BORING FACT TO SELL

> **“No artificial preservatives.” Every brand says it; nobody believes it. So they photographed one Whopper going mouldy for over a month.**

The Moldy Whopper won the Cannes Lions Grand Prix. Can AI find that kind of idea in data — before a creative team does?

LECTURE 9 · WHAT YOU'LL LEARN

Turn audience data into one message worth paying to show.

PERSONA

Who are we talking to?

Skeptical of “clean” claims

→

TENSION

What's bugging them?

Clean-label fatigue: no proof

→

ANGLE

Why does that happen?

Costly signalling

→

STORY

What's the one line?

“The beauty of no artificial preservatives”

THREE PHILOSOPHIES FOR AI IN MARKETING

- ✓ **Replacement:** AI runs creative and targeting end to end (Meta's vision)
- ✓ **Empowerment:** AI co-creates with people (Google's ACAI)
- ✓ **Insight:** AI finds patterns that guide decisions (SOMIN, Nielsen)

REAL CASE · INDRIVE PHILIPPINES

169 → 108

Google ads analysed → creative expressions generated.
The gap found: Safety & Trust, unclaimed by inDrive.

SOURCE Farseev, “From Signal to Story to Spend”, Future-Proof Forum '26, Google HQ Manila; Farseev, “Will AI Make Agencies Obsolete?”, ACM MM '25 talk; ITMO SMA (influencers).

10

LECTURE 10

Business applications

WHEN TWEETS MOVE MONEY

> **Twitter mood predicted the direction of the Dow Jones with 86.7% accuracy.**

Bollen et al., 9.7M tweets, as covered in the ITMO course. Beyond marketing, the same signals feed customer intelligence, decision support and automated workflows across a company.

LECTURE 10 · WHAT YOU'LL LEARN

The same technology, pointed at the whole business.

CUSTOMER INTELLIGENCE

Miraflora Wagyu

High acquisition costs looked like a pricing problem. Audience data said otherwise:

“They're not price-sensitive — they're deal-sensitive.”

WORKFLOW AUTOMATION

Creator commerce, UK

1,047 creators on the roster, 18 activated, 6 selling. Briefing took ~40 hours per cycle.

AI scoring and briefs: 75% lower cost (SOMIN case).

ORGANISATIONAL CHANGE

Agencies & teams

Repetitive work (resizing, reporting, A/B tests) is being automated; new roles are emerging.

WPP ran 150,000+ AI training sessions in 2025.

✓ **Your takeaway:** judge any AI system by the decision it improves, what it costs, and who stays accountable.

SOURCE ITMO SMA — Sentiment Analysis (Bollen et al.); Farseev, DMC KL '25 talk (SOMIN × GWI Core Q2 2024); SOMIN creator-commerce pitch, Sep 2026; ACM MM '25 talk.

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BY THE END OF THE COURSE

What you'll be able to do.

- ✓ **Explain** how user profiling works, and where it goes too far.
- ✓ **Analyse** networks: influence, virality and fake accounts.
- ✓ **Connect** classic retrieval to modern AI search and RAG.
- ✓ **Build** and compare recommendation approaches.
- ✓ **Choose** the right metric, not just accuracy.
- ✓ **Reason** about what LLMs can and can't do for a business.
- ✓ **Design** an agentic workflow and name its risks.
- ✓ **Apply** it all to targeting, personalisation and campaigns.
- ✓ **Ship** a tech-enabled product — and defend it.

BEFORE LECTURE 1 · OPTIONAL

Watch or read ahead.

01	**INFORMATION RETRIEVAL**	Stanford CS276 — Scoring, term weighting & the vector space model web.stanford.edu/class/cs276/19handouts/lecture6-tfidf-1per.pdf
02	**RECOMMENDER SYSTEMS**	J. Leskovec, Stanford CS246 — Recommender Systems I web.stanford.edu/class/cs246/slides/07-recsys1.pdf
03	**LARGE LANGUAGE MODELS**	Y. Dubois, Stanford CS229 — Introduction to Building LLMs (2024) www.youtube.com/watch?v=9vM4p9NN0Ts
04	**AGENTIC AI**	K. Katanforoosh, Stanford CS230 — Agents, Prompts, and RAG (2025) www.youtube.com/watch?v=k1njvbBmfsw
05	**AI IN MARKETING, IN PRACTICE**	Google Cloud customer story — SOMIN.AI cloud.google.com/customers/somin-ai

QUESTIONS

One course. One project.
One question: how does data about people become a business decision?

DeepMind. Keeta. The next alumni story could be yours.

See you in Lecture 1: Introduction to social media analytics.